



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2009
YEAR 12 Mathematics Extension 1
HSC Task #3

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each Section is to be returned in a separate bundle.
- All answers must be given in exact simplified form unless otherwise stated.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.

Total Marks – 69

- Attempt questions 1-3
- Start each new section in a separate answer booklet

Examiner: *D.McQuillan*

Question 1 (24 marks)**Marks**

(a) Differentiate

(i) e^{2x} **1**

(ii) $\ln(2x - 5)$ **1**

(iii) $x^2 e^{4x}$ **2**

(iv) $\frac{\ln x}{x}$ **2**

(b) Find

(i) $\int e^{1-x} dx$ **1**

(ii) $\int \frac{dx}{2x - 7}$ **1**

(c) Evaluate

(i) $\int_0^1 e^{4x} dx$ **2**

(ii) $\int_3^7 \frac{3x}{x^2 - 5} dx$ **2**

(d) Evaluate

(i) $\tan^{-1}(\sqrt{3})$ **1**

(ii) $\sin^{-1}\left(\sin \frac{\pi}{4}\right)$ **1**

(iii) $\cos\left(2 \sin^{-1}\left(\frac{5}{13}\right)\right)$ **2**

(e) Using the substitution $u = x^2$ find $\int 2xe^{x^2} dx$. **2**

(f) Find the equation of the tangent to the curve $y = e^{3x-1}$ at the point where $x = 2$. **3**

(g) Solve the following pair of equations simultaneously. **3**

$$\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$$

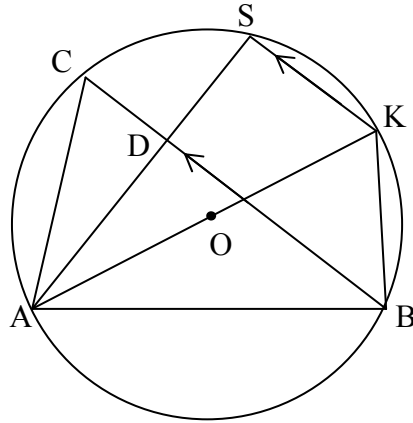
$$\sin^{-1} x - \sin^{-1} y = \frac{\pi}{6}$$

End of Question 1

Question 2 (24 marks)

Marks

- (a) AK is a diameter of the circle, centre O. SK||CB.
- (i) Prove that $AS \perp BC$. **2**
- (ii) Let $\angle BAK = \alpha$, hence show that $\angle SAC = \alpha$. **2**



- (b)
- (i) Show that $u^2 + u + 1 + \frac{1}{u-1} = \frac{u^3}{u-1}$. **1**
- (ii) Hence find $\int \frac{dx}{\sqrt{x} - \sqrt[3]{x}}$ using the substitution $x = u^6$. **2**
- (c) The curve $y = \frac{3}{\sqrt{x^2 + 4}}$ is rotated about the x-axis between $x = 0$ and $x = 2$. Find the volume of the solid generated. **3**
- (d) Given that $f(x) = 2 - \frac{1}{2e^{x^3}}$
- (i) show that $f'(x) + 3x^2 f(x) = 6x^2$ **2**
- (ii) show that the continuous function $f(x)$ has a root between $x = -2$ and $x = -1$ **1**
- (iii) use one application of Newton's Method with initial approximation of $x = -1$ to find an approximation to the root of $f(x)$. Give your answer to 3 decimal places. **2**

(e) Evaluate $\int_{-3}^0 \frac{x^2}{\sqrt{1-x}} dx$ using the substitution $u = 1 - x$. **3**

(f) Let $y = \cot^{-1} x$ be defined as $x = \cot y$ for $0 < y < \frac{\pi}{2}$.

(i) Graph $y = \cot^{-1} x$. **1**

(ii) Show that $\frac{dy}{dx} = -\frac{1}{1+x^2}$. **2**

(iii) Hence or otherwise show that, for all $x > 0$,

$$f(x) = \tan^{-1} x + \cot^{-1} x$$

is a constant function. **2**

(iv) Find the value of $\tan^{-1} x + \cot^{-1} x$ in exact form. **1**

End of Question 2

Question 3 (21 marks)

Marks

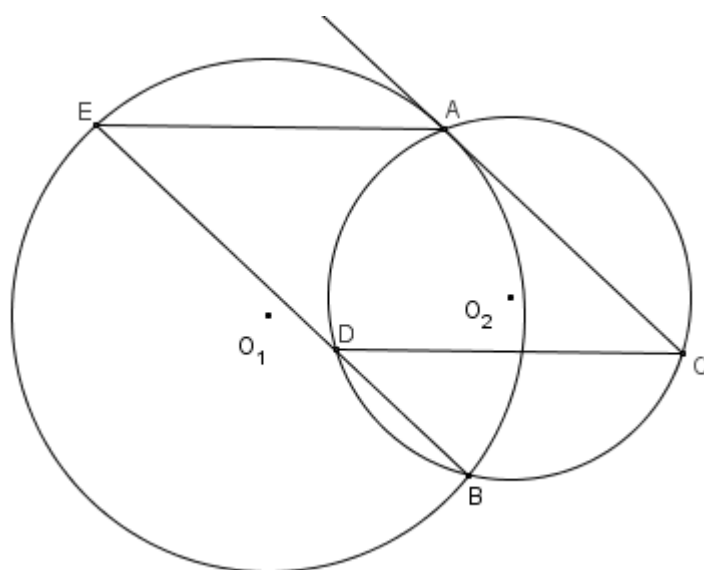
- (a) Let $I = \int_0^{\pi} xf'(\sin x)dx$. Using the substitution $u = \pi - x$ show that

$$2I = \pi \int_0^{\pi} f(\sin x)dx.$$

3

- (b) Two circles O_1 and O_2 meet at the points A and B. When produced, the tangent to O_1 at A meets O_2 at C. A point E lies on the circumference of O_1 so that BE is parallel to CA. The chord BE meets O_2 at D. Prove that $AE = CD$.

3

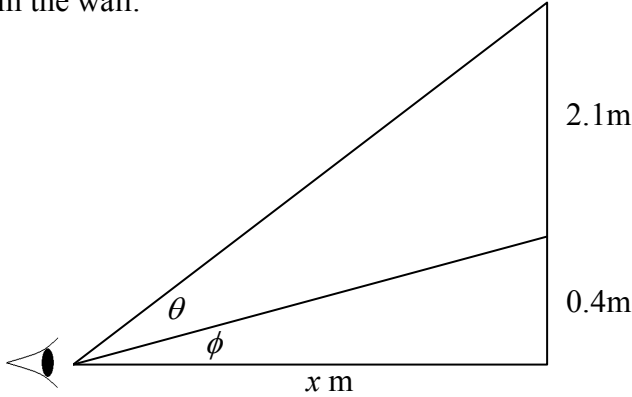


- (c) Using Mathematical Induction, prove that for any integer $n \geq 1$,

$$5 \times 2^{3n-2} + 3^{3n-1}$$

is divisible by 19.

3

- (d) Given that $f(x) = \frac{e^x - e^{-x}}{2}$.
- (i) Show that $f^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$ **3**
- (ii) Solve the equation $f(x) = 5$ to 2 decimal places. **1**
- (iii) Find $\frac{d}{dx}[f^{-1}(x)]$. **2**
- (e) A painting in an art gallery has height 2.1 m and is hung so that its lower edge is 0.4 metres above the eye of an observer who is standing x metres away from the wall.
- 
- (i) Find an expression for ϕ in terms of x . **1**
- (ii) Hence, find θ as a function of x . **2**
- (iii) How far from the wall should the observer stand to get the best view? (That is, find x such as to maximise the viewing angle θ .) **3**

End of Question 3

End of Exam

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

QUESTION 1

(a) (i) $2e^{2x}$

(ii) $\frac{2}{2x-5}$

(iii) $2xe^{4x} + 4x^2e^{4x}$

(iv) $\left(\frac{1}{x} \ln x\right)' = -x^{-2} \ln x + \frac{1}{x^2}$
 $= \frac{1}{x^2} - \frac{\ln x}{x^2}$

(b) (i) $\int e^{1-x} dx = -e^{1-x} + C$

(ii) $\int \frac{dx}{2x-7} = \frac{1}{2} \int \frac{2dx}{2x-7}$

$$= \frac{1}{2} \ln(2x-7) + C$$

(c) (i) $\int_0^1 e^{4x} dx = \left. \frac{1}{4} e^{4x} \right|_0^1$
 $= \frac{e^4}{4} - \frac{1}{4}$

(ii) $\int_3^7 \frac{3x}{x^2-5} dx = \frac{3}{2} \int_3^7 \frac{2x}{x^2-5} dx$

$$= \frac{3}{2} \left[\ln(x^2-5) \right]_3^7$$

$$= \frac{3}{2} (\ln 44 - \ln 4)$$

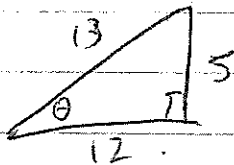
$$= \frac{3}{2} \ln 11$$

$$(d) (i) \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$(ii) \sin^{-1}\left(\sin \frac{\pi}{4}\right) = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$= \frac{\pi}{4}$$

$$(iii) \cos\left(2\sin^{-1}\left(\frac{5}{13}\right)\right) = \cos(2\theta)$$



$$= \cos^2 \theta - \sin^2 \theta$$

$$= \left(\frac{12}{13}\right)^2 - \left(\frac{5}{13}\right)^2$$

$$= \frac{144 - 25}{169}$$

$$= \frac{119}{169}$$

$$(e) \int 2x e^{x^2} dx$$

$$= \int e^u du$$

$$= e^u + C$$

$$= e^{x^2} + C$$

$$u = x^2$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

$$(f) y = e^{3x-1} \quad x=2$$

$$\frac{dy}{dx} = 3e^{3x-1}$$

$$\text{Grad at } x=2$$

$$m = 3e^5$$

$$(2, e^5)$$

$$y - e^5 = 3e^5(x - 2)$$

$$y = 3e^5 x - 5e^5$$

$$(9) \quad \begin{aligned} \sin^{-1}x + \sin^{-1}y &= \frac{\pi}{3} & \textcircled{A} \\ \sin^{-1}x - \sin^{-1}y &= \frac{\pi}{6} & \textcircled{B} \end{aligned}$$

$$\textcircled{A} + \textcircled{B}$$

$$2\sin^{-1}x = \frac{2\pi}{3}$$

$$\sin^{-1}x = \frac{\pi}{3}$$

$$x = \frac{\sqrt{3}}{2}$$

$$\textcircled{A} - \textcircled{B}$$

$$2\sin^{-1}y = \frac{\pi}{3}$$

$$\sin^{-1}y = \frac{\pi}{6}$$

$$y = \frac{1}{2}$$

$$x = \frac{\sqrt{3}}{2}$$

2) a) i) $\angle ASK = 90^\circ$ (angle in semi circle)

$\angle ADB = 90^\circ$ (corr. $\angle$'s $CB \parallel SK$)

$\therefore AS \perp BC$

ii) let $\angle BAK = \alpha$

$\angle ABK = 90^\circ$ (angle in semi circle)

$\angle BKA = 90 - \alpha$ ($\angle$ sum of Δ)

$\angle ACB = 90 - \alpha$ ($\angle$ in same segment)

$\angle ADC = 90^\circ$ ($AS \perp BC$)

$\angle SAC = \alpha$ ($\angle$ sum of Δ)

b) i) LHS = $u^2 + u + 1 + \frac{1}{u-1}$

$= \frac{(u-1)(u^2+u+1) + 1}{u-1}$

$= \frac{u^3 - 1 + 1}{u-1}$

$= \frac{u^3}{u-1}$

$= \text{RHS}$

ii) $\int \frac{dx}{\sqrt{x} - \sqrt[3]{x}}$

$x = u^6$

$u = x^{\frac{1}{6}}$

$\frac{dx}{du} = 6u^5$

$dx = 6u^5 du$

$= \int \frac{6u^5 du}{u^3 - u^2}$

$= \int \frac{6u^3}{u-1} du$

$= 6 \int (u^2 + u + 1 + \frac{1}{u-1}) du$

$= 6 \left[\frac{u^3}{3} + \frac{u^2}{2} + u + \ln(u-1) \right] + C$

$= 6 \left[\frac{\sqrt{x}}{3} + \frac{\sqrt[3]{x}}{2} + \sqrt[6]{x} + \ln(\sqrt[6]{x} - 1) \right] + C$

$$c) V = \pi \int_a^b y^2 dx$$

$$V = \pi \int_0^2 \frac{9}{x^2+4} dx$$

$$V = 9\pi \int_0^2 \frac{dx}{x^2+4}$$

$$V = 9\pi \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^2$$

$$V = 9\pi \left[\frac{1}{2} \tan^{-1} \frac{2}{2} - \cancel{\frac{1}{2} \tan^{-1} \frac{0}{2}} \right]$$

$$V = 9\pi \left[\frac{1}{2} \times \frac{\pi}{4} \right]$$

$$V = \frac{9\pi^2}{8} \text{ units}^3$$

$$d) i) f(x) = 2 - \frac{1}{2e^{x^3}}$$

$$= 2 - \frac{1}{2} e^{-x^3}$$

$$f'(x) = \frac{3x^2}{2} e^{-x^3}$$

$$f'(x) + 3x^2 f(x) = \frac{3x^2}{2} e^{-x^3} + 3x^2 \left(2 - \frac{1}{2} e^{-x^3} \right)$$

$$= \cancel{\frac{3x^2}{2} e^{-x^3}} + 6x^2 - \cancel{\frac{3x^2}{2} e^{-x^3}}$$

$$= 6x^2$$

$$ii) f(-2) = 2 - \frac{1}{2} e^{-(-2)^3}$$

$$= -1488.48 \dots$$

$$< 0$$

$$f(-1) = 2 - \frac{1}{2} e^{-(-1)^3}$$

$$= 0.640859 \dots$$

$$> 0$$

since the continuous function $f(x)$ has a change in sign at least one root lies between $x = -2$ & $x = -1$.

$$\text{iii) } a_1 = a_0 - \frac{f(a_0)}{f'(a_0)}$$

$$f(-1) = 0.640859085$$

$$f'(-1) = \frac{3(-1)^2}{2} e^{-(-1)^3}$$

$$= 4.077422743$$

$$a_1 = -1 - \frac{0.640859085}{4.077422743}$$

$$a_1 = -1.157 \text{ (3 dec. places)}$$

$$\text{e) } \int_{-3}^0 \frac{x^2}{\sqrt{1-x}} dx$$

$$u = 1 - x$$

$$x = 1 - u$$

$$\frac{du}{dx} = -1$$

$$= \int_4^1 \frac{(1-u)^2}{\sqrt{u}} - du$$

$$dx = -du$$

$$\text{when } x = 0$$

$$x = -3$$

$$u = 1$$

$$u = 4$$

$$= \int_1^4 \frac{1 - 2u + u^2}{\sqrt{u}} du$$

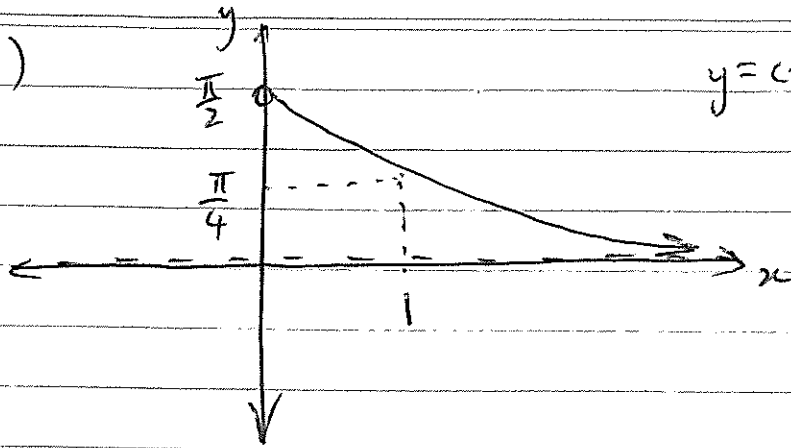
$$= \int_1^4 (u^{-\frac{1}{2}} - 2u^{\frac{1}{2}} + u^{\frac{3}{2}}) du$$

$$= \left[2u^{\frac{1}{2}} - \frac{4}{3}u^{\frac{3}{2}} + \frac{2}{5}u^{\frac{5}{2}} \right]_1^4$$

$$= \left[2(4)^{\frac{1}{2}} - \frac{4}{3}(4)^{\frac{3}{2}} + \frac{2}{5}(4)^{\frac{5}{2}} - \left(2 - \frac{4}{3} + \frac{2}{5} \right) \right]$$

$$= \frac{76}{15}$$

f) i)



$$y = \cot^{-1} x \rightarrow x = \cot y$$

$$0 < y < \frac{\pi}{2}$$

ii) $y = \cot^{-1} x$

$$x = \cot y$$

OR $x = \frac{1}{\tan y}$

$$\frac{dx}{dy} = -\operatorname{cosec}^2 y$$

$$\tan y = \frac{1}{x}$$

$$\frac{dy}{dx} = -\frac{1}{\operatorname{cosec}^2 y}$$

$$y = \tan^{-1}\left(\frac{1}{x}\right)$$

$$= -\frac{1}{1 + \cot^2 y}$$

$$\frac{dy}{dx} = \frac{1}{1 + \left(\frac{1}{x}\right)^2} \cdot -\frac{1}{x^2}$$

$$= -\frac{1}{1 + x^2}$$

$$= -\frac{1}{1 + x^2}$$

iii) $f(x) = \tan^{-1} x + \cot^{-1} x$

$$f'(x) = \frac{1}{1+x^2} + -\frac{1}{1+x^2}$$

$$= 0$$

$\therefore f(x)$ is a constant function

iv) $f(1) = \tan^{-1} 1 + \cot^{-1} 1$

$$= \frac{\pi}{4} + \frac{\pi}{4}$$

$$= \frac{\pi}{2}$$

$$\therefore f(x) = \frac{\pi}{2}$$

Question 3

$$(a) \quad I = \int_0^{\pi} x f(\sin x) dx$$

Let $u = \pi - x$
 $\therefore du = -dx$

When $x = 0$, $u = \pi$
 $x = \pi$, $u = 0$

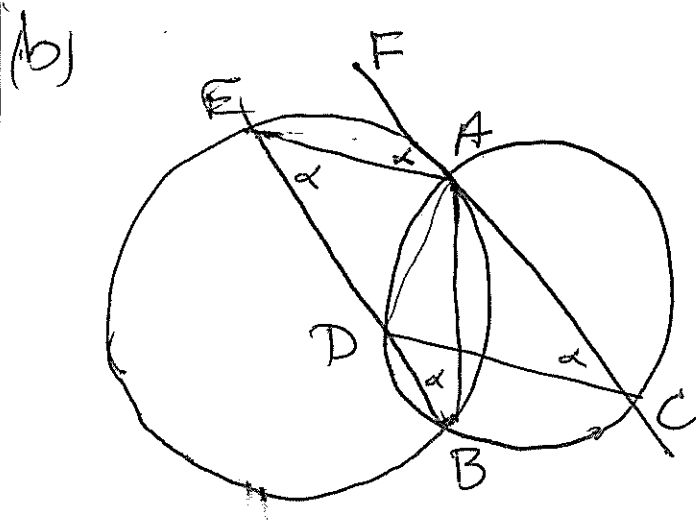
$$\therefore I = \int_{\pi}^0 (\pi - u) f(\sin(\pi - u)) (-du)$$

$$= \int_0^\pi \pi f(\sin u) du - \int_0^\pi u f(\sin u) du$$

$$= \int_0^\pi \pi f(\sin x) dx - \int_0^\pi x f(\sin x) dx \quad (*)$$

$$\therefore 2I = \int_0^\pi \pi f(\sin x) dx \quad [3]$$

(*) The variable name is a dummy - swap u for x .



Join AB

Let $\angle FAE = \alpha$

$$\angle AEB = \alpha \text{ (alternate } \angle\text{'s, } AE \parallel CD)$$

$\angle ABE = \alpha$ (alternate segment)

$$\angle ACD = \alpha \text{ (standing on same chord)}$$

ie $\angle FAE = \angle ACD$ and so

$AE \parallel DC$ (Corres p. angles)

$\therefore$ EACB is a parallelogram
(a pair of sides equal & parallel)

$\therefore EA = DC$ (Opp sides of parallelogram equal)
QED

[3]

(C) $p(n)$: Aim to prove $19 \mid 5 \times 2^{3n-2} + 3^{3n-1}$

$p(i)$: Test when $n=1$

$$5x^{2^{3-2}} + 3^{3-1}$$

$\equiv 19 \quad [1]$

P(12): Assume true for $n=k$

$$5 \times 2^{3k-2} + 3^{3k-1} = 19R$$

$p(k+1)$: RTP such that $p(k) \rightarrow p(k+1)$

$$5 \times 2^{3k+1} + 3^{3k+2}$$

$$= 40 \times 2^{3k-2} + 27(198 - 5 \times 2^{3k-2})$$

$$= 40 \times 2^{3k-2} + 27 \times 192 - 135 \times 2^{3k-2}$$

$$= 27 \times 19R - 95 \times Z^{3k-2}$$

$$= 19(27R - 5 \times 2^{3x-2}) \quad [2]$$

$$\therefore p(k) \rightarrow p(k+1).$$

Thus $p(n)$ is true for $n \geq 1$.

$$(d) f(x) = \frac{e^x - e^{-x}}{2}$$

(i) Invert by exchanging x and y .

$$x = \frac{e^y - e^{-y}}{2}$$

$$2x = e^y - e^{-y}$$

$$2xe^y = e^{2y} - 1 \quad (\text{multiply by } e^y)$$

$$e^{2y} - 2xe^y - 1 = 0 \quad (\text{quadratic})$$

$$\therefore e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2}$$

$$e^y = x \pm \sqrt{x^2 + 1}$$

Take logs, drop spurious -ve

$$y = \ln(x + \sqrt{x^2 + 1}) \quad [3]$$

$$(ii) 5 = \frac{e^x - e^{-x}}{2}$$

$$\text{So } e^{2x} - 10e^x - 1 = 0$$

$$e^x = \frac{10 + \sqrt{100 + 4}}{2}$$

$$= 10.099...$$

$$x \doteq 2.31 \quad [1]$$

$$(iii) \frac{d}{dx} \left(\ln(x + \sqrt{x^2 + 1}) \right)$$

$$= \frac{1}{x + \sqrt{x^2 + 1}} \left(1 + \frac{x}{\sqrt{x^2 + 1}} \right)$$

$$= \frac{1}{x + \sqrt{x^2 + 1}} \left(\frac{x + \sqrt{x^2 + 1}}{\sqrt{x^2 + 1}} \right)$$

$$= \frac{1}{\sqrt{x^2 + 1}} \quad [2]$$

$$(e)(i) \tan \phi = \frac{0.4}{x}$$

$$\therefore \phi = \tan^{-1} \frac{0.4}{x} \quad [1]$$

$$(ii) \theta + \phi = \tan^{-1} \left(\frac{2.5}{x} \right)$$

$$\therefore \theta = \tan^{-1} \left(\frac{2.5}{x} \right) - \tan^{-1} \left(\frac{0.4}{x} \right) \quad [2]$$

$$(iii) \frac{d\theta}{dx} = \frac{-2.5}{x^2 + 6.25} + \frac{0.4}{x^2 + 0.16}$$

$$= \frac{-2.5(x^2 + 0.16) + 0.4(x^2 + 6.25)}{(x^2 + 6.25)(x^2 + 0.16)}$$

$$\frac{d\theta}{dx} = 0 \quad \text{when}$$

$$-2.5(x^2 + 0.16) + 0.4(x^2 + 6.25) = 0$$

$$2.1(1 - x^2) = 0$$

$$1 - x^2 = 0$$

$$\therefore x = \pm 1$$

$$\text{But } x > 0$$

$$\therefore x = 1 \quad [3]$$